10/7 박수민 7 bit full adder

1. simulator
[source code]
```
`timescale 1ns / 1ps

module tb_adder7();

    reg   [6:0] a;
    reg   [6:0] b;
    reg         cin;

    wire [6:0] sum;
    wire        cout;


    adder7 u_adder7(
        .a(a),
        .b(b),
        .cin(cin),
        .sum(sum),
        .cout(cout)
    );


    initial begin

        // 1 + 1 = 2
        a = 7'b0000001;
        b = 7'b0000001;
        cin = 1'b0;
        #20;

        // 3 + 5 = 8
        a = 7'b0000011;
        b = 7'b0000101;
        cin = 1'b0;
        #20;

        // 7 + 9 = 16
```

```
        a = 7'b0000111;
        b = 7'b0001001;
        cin = 1'b0;
        #20;

        // 63 + 1 = 64
        a = 7'b0111111;
        b = 7'b0000001;
        cin = 1'b0;
        #20;

        // 127 + 1 = 128
        a = 7'b1111111;
        b = 7'b0000001;
        cin = 1'b0;
        #20;

        // cin 동작 확인
        // 1 + 1 + 1 = 3
        a = 7'b0000001;
        b = 7'b0000001;
        cin = 1'b1;
        #20;

        // 최대값 확인
        // 127 + 127 + 1 = 255
        a = 7'b1111111;
        b = 7'b1111111;
        cin = 1'b1;
        #20;

        $stop;

    end

endmodule
```

[timing chart]



2. basys3에서의 동작 확인 결과
[source code]
```
adder7.v
`timescale 1ns / 1ps

module adder7(

    input [6:0] a,
    input [6:0] b,
    input cin,
    output [6:0] sum,
    output cout

    );

    wire w_c0, w_c1, w_c2, w_c3, w_c4, w_c5;

    full_adder u0_full_adder(
        .a(a[0]),
        .b(b[0]),
        .cin(cin),
        .sum(sum[0]),
        .cout(w_c0)
    );

    full_adder u1_full_adder(
        .a(a[1]),
        .b(b[1]),
        .cin(w_c0),
        .sum(sum[1]),
        .cout(w_c1)
    );

    full_adder u2_full_adder(
        .a(a[2]),
        .b(b[2]),
        .cin(w_c1),
        .sum(sum[2]),
        .cout(w_c2)
```

```
    );

    full_adder u3_full_adder(
        .a(a[3]),
        .b(b[3]),
        .cin(w_c2),
        .sum(sum[3]),
        .cout(w_c3)
    );

    full_adder u4_full_adder(
        .a(a[4]),
        .b(b[4]),
        .cin(w_c3),
        .sum(sum[4]),
        .cout(w_c4)
    );

    full_adder u5_full_adder(
        .a(a[5]),
        .b(b[5]),
        .cin(w_c4),
        .sum(sum[5]),
        .cout(w_c5)
    );

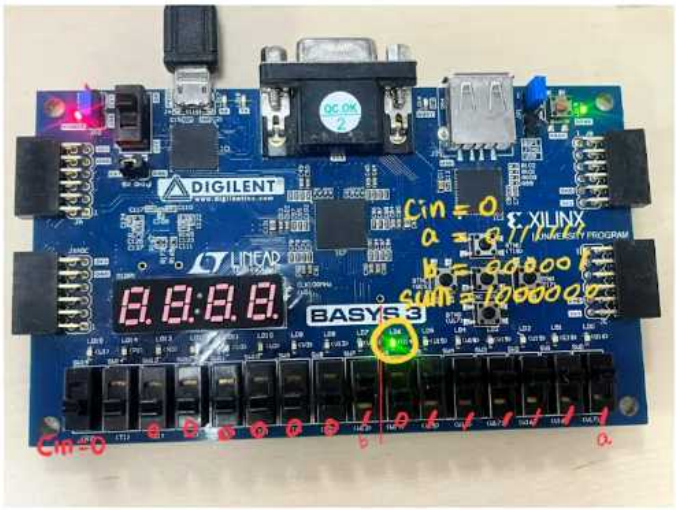
    full_adder u6_full_adder(
        .a(a[6]),
        .b(b[6]),
        .cin(w_c5),
        .sum(sum[6]),
        .cout(cout)
    );

endmodule
```

[동작 결과]

basys3 동작 결과

1) a=0111111 , b= 0000001, Cin = 0 → Sum = 1000000



2) a = 0000001 , b= 0000001, Cin = 1 → Sum = 0000011

